ABDUR REHMAN AFZAL

Senior AI Engineer | LLM Agents · RAG · Voice AI · Computer Vision | Senior Python Engineer
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Lahore, Pakistan | Remote | Open to relocation: GCC, Europe, North America

PROFESSIONAL SUMMARY

Senior AI Engineer at MindMeta AI (UK, remote) with 7+ years building production software, the last 2+ focused on shipping LLM, agentic and voice AI systems to production: a LangGraph + Vision-Language Model annotation pipeline that auto-labels dashcam safety events for a fleet-safety telematics platform (Kubernetes, US/EU/Canada), a LiveKit SIP voice concierge running on hotel phone systems for the Saudi market, and RAG assistants serving Dubai real estate. Strong Python backend foundation (FastAPI, Django, PostgreSQL, AWS/GCP, Kubernetes). Measured impact: 60% higher lead qualification, ~70% less manual document processing, 30% lower cloud cost. GCC client experience across UAE and Saudi Arabia, including Arabic/English products.

TECHNICAL SKILLS

LLM & Agents: LangChain, LangGraph, CrewAI, OpenAI API (GPT-4o / GPT-4), OpenRouter, Groq, Function / Tool Calling, Structured Outputs (Pydantic), Prompt Engineering, Guardrails, RAG Evaluation, Multi-Agent Systems, n8n Automation

RAG & Search: Retrieval-Augmented Generation, Embeddings (Sentence-Transformers), ChromaDB, Pinecone, pgvector, Hybrid Search (vector + keyword / pg_trgm), Semantic Search, Snowflake, BigQuery

Voice AI & Telephony: LiveKit Agents & SIP, Pipecat, WebRTC, Deepgram STT / TTS, Cartesia, Silero VAD, Turn Detection, SIP / VoIP, PBX Integration (Yeastar, 3CX), DTMF, G.722

Computer Vision & ML: Vision-Language Models (VLMs), YOLOv8, PyTorch, OpenCV, Tesseract / EasyOCR, NVIDIA Triton Inference Server, AI Data Annotation / Auto-Labeling, Statsig Dynamic Config

Backend & APIs: Python 3.12, FastAPI, Django / DRF, Django Channels, Celery, Huey, PostgreSQL, Redis, GraphQL, REST, WebSockets, gRPC, OAuth 2.0 / JWT, Node.js

Cloud, MLOps & DevOps: AWS (S3, EC2, RDS, ECS, ECR, SQS, SES), GCP (GKE, BigQuery), Docker, Kubernetes, ArgoCD (GitOps), Terraform, Datadog, GitHub Actions CI/CD, Nginx, Linux

Frontend: React 19, Next.js 16, TypeScript, TailwindCSS, TanStack Query, i18n / RTL (Arabic & English), Vitest, Playwright, Angular

PROFESSIONAL EXPERIENCE

MindMeta AI | United Kingdom (Remote) — *Mar 2025 – Present*

Senior AI Engineer (Contract) — UK AI company delivering agentic AI, voice AI and AI search products for clients in the US and Saudi Arabia

AI Annotation service for a fleet-safety telematics platform (US) — production

- Own the ML/AI side of a production annotation service that automatically reviews and labels dashcam safety events (crashes, hard braking, seat-belt violations, camera obstruction), replacing slow manual review at scale.
- Designed the annotation pipelines and built per-event agentic workflows in LangChain / LangGraph that combine Triton-served computer vision models with multi-step VLM reasoning (OpenRouter); implemented false-positive rejection and Statsig-driven threshold tuning to keep labels accurate.
- Integrated AWS SQS intake, Snowflake data and Triton inference; ships on Kubernetes across US, EU and Canada via ArgoCD, Terraform, Datadog and GitHub Actions CI/CD.

Voicecake · Mojeeb AI (Saudi Arabia) — multi-tenant white-labelled AI agent platform, mojeebai.com — production pilot

- Built the voice channel: a LiveKit SIP agent (Deepgram STT/TTS, Groq LLM, Silero VAD) on the hotel's own Yeastar PBX — guests dial 800 from the room phone; automatic room identification, SIP REFER transfer to reception in ~0.6 s, PBX-originated wake-up calls (agent speaks within 1.3 s); validated G.722, DTMF and concurrent calls.
- Front-end developer of the owner web app (Next.js 16, React 19, TypeScript): conversations, Yes / No / Edit approvals, activity with proof, rulebook, live SSE updates; fully bilingual Arabic / English RTL; OpenAPI contract-first, Vitest + Playwright in CI.

Nightlife.net (US) — AI Search Assistant (in development)

- Designing a standalone FastAPI search assistant for a US nightlife directory: hybrid retrieval (pgvector + pg_trgm), hard SQL filters for city / date / category extracted from the query, GPT-4o-mini answers grounded only in retrieved listings, Redis caching and a nightly re-embedding sync.

B1 Properties | Dubai, UAE (Remote) — *Nov 2024 – Oct 2026*

Full-Stack Python & AI Developer

- Increased lead qualification rate by 60% with a LangChain + RAG + ChromaDB real-estate chatbot (GPT-4, Telegram Bot API); cut daily manual reporting by 70% with an n8n + GPT-4 automation pipeline (~2 hours/day reclaimed).
- Delivered b1properties.ae end to end: Django 5.2 / DRF (50+ REST endpoints), Next.js 16 frontend, hardened Nginx / Gunicorn / PM2 stack; eliminated N+1 queries for a 30% API speed-up and zero production downtime since launch.

Aircod Technologies | Lahore, Pakistan *2024 – 2025*
Senior Python Backend & AI Developer

- Cut document processing turnaround ~70% across 3+ client pipelines with LangChain + CrewAI + OpenAI multi-agent document triage.
- Built an AI document extraction system for Hamriyah Free Zone Authority (UAE government): YOLOv8 + PyTorch detection, dual OCR (Tesseract + EasyOCR), ICAO 9303 MRZ parsing — 40+ fields from Emirates IDs, passports and visas, processing time from minutes to seconds.
- Contributed to AdsGency AI (150+ businesses, $50M+ managed ad spend): OpenAI-powered ad-copy generation module, Google Ads and Meta Marketing API integrations, FastAPI / DRF analytics APIs.
- Reduced cloud costs 30% by migrating monolithic Django to FastAPI microservices on AWS; Docker + GitHub Actions CI/CD lifted API throughput 35% and halved release cycle time.

Synares Systems | Lahore, Pakistan *2023 – 2024*
Senior Software Engineer II / Backend Team Lead

- Led 5 engineers delivering TimeKeepers / WorkFly (workfly.app), a multi-tenant workforce SaaS: Django Channels + WebSockets + Redis real-time layer (sub-100 ms), GPS geofenced time-clock, AWS ECS CI/CD with deploys under 5 minutes; code-review cycle time down ~40%.

Merik Solutions | Islamabad, Pakistan *2022 – 2023*
Full-Stack Python Developer

- Cut production bug rate 30% across 3+ enterprise Django applications by introducing PyTest / TDD from zero coverage; delivered Wagtail CMS platforms, DRF APIs and automated operational dashboards.

Viral Square | Lahore, Pakistan *2021 – 2022*
Python Developer

- Built Palletfly (dbr.palletfly.com), a B2B supply-chain platform: 40,000+ lines of Python across 10 Django apps, real-time collaborative inventory editing (Channels + Redis), Huey async jobs cutting bulk operations ~90%.

UEnergy Solar | Huddersfield, UK (Remote) *2018 – 2021*
Software Engineer

- Built and maintained two production platforms for a UK renewable-energy company serving BP, Shell, NHS and Sainsbury's: Django CRM and quoting system (~60% faster proposals), automated billing, and energy analytics dashboards.

University of Central Punjab | CS Instructor (Part-Time), 2018 – 2021: taught 200+ undergraduates across 7 Computer Science courses (OOP, DSA, Operating Systems, Networks).

SELECTED AI PROJECTS

Portfolio RAG Assistant & Real-Time Voice Agent | abdurrehmanafzal.cloud · talk.abdurrehmanafzal.cloud
Tech: *FastAPI, OpenAI GPT-4o-mini, Sentence-Transformers, ChromaDB, Pydantic, Pipecat, WebRTC, Deepgram, Cartesia, Silero VAD, Docker, GitHub Actions*

- RAG assistant with OpenAI tool calling, Pydantic structured-output lead capture, prompt-injection guardrails and a keyword-based RAG evaluation harness; multi-turn session memory.
- Interruptible browser voice agent (WebRTC → VAD → Deepgram STT → GPT-4o-mini → Cartesia TTS) with barge-in and RAG via tool calling; Dockerized behind Traefik with GitHub Actions auto-deploy and cost guardrails.

AI-Assisted News Intelligence Platform
Tech: *Node.js, Express, PostgreSQL, OpenAI GPT-4, React, node-cron, PDFKit*

- End-to-end pipeline: multi-source RSS ingestion → weighted keyword filtering → GPT-4 summarisation with a dynamic prompt-templating engine → PDF reports → scheduled email delivery; 7-day deduplication window; 12-component React dashboard.

EDUCATION & CERTIFICATIONS

Bachelor of Science in Computer Science — University of Central Punjab (UCP), Lahore, Pakistan *June 2018*
Certifications: The AI Engineer Course 2025: Complete AI Engineer Bootcamp (Udemy) · AI Engineer Agentic Track: The Complete Agent & MCP Course (Udemy) · n8n: AI Agents, AI Automations & AI Voice Agents (Udemy) · Python, Cloud & AI (Coursera) · Full Stack Developer Certification (EVS Training Institute)
Languages: English (professional), Urdu (native)